14

The Spectral Theorem

Historical Note

This is the oldest theorem in the book and the last one we prove. Cauchy showed in 1829 that the roots of the secular equation of a symmetric array are real and that its principal axes are perpendicular, working entirely in the language of quadratic forms and rigid-body mechanics; Chapter 9 quoted that result and deferred it here. Sylvester added the law of inertia in 1852, which is Chapter 15. The proof we give is not Cauchy's: it goes through a triangularisation that Issai Schur, a student of Frobenius, published in 1909, and which turns out to make the symmetric case almost immediate. Toeplitz identified in 1918 exactly which matrices the argument reaches, namely the normal ones. And the infinite-dimensional version, which is what quantum mechanics actually needs, was assembled by Hilbert and completed by von Neumann around 1930, a full century after Cauchy. Few theorems have been proved so many times by so many good people.

Three chapters have now promised this result. Chapter 9 promised that a symmetric matrix has real eigenvalues, Chapter 10 promised that it is always diagonalisable and by an orthonormal basis, and both promised Schur's triangularisation once the word *unitary* was available. Chapter 12 supplied the word. This chapter pays all three debts, and the tool that does it is the adjoint.

14.1 Adjoints

Proposition 14.1 (The adjoint of a matrix)**.** *For $A \in M_{m\times n}(\mathbb{C})$ and the standard inner products,*

$$\langle Ax, y\rangle = \langle x, A^{*}y\rangle \qquad \text{for all } x \in \mathbb{C}^{n},\ y \in \mathbb{C}^{m},$$

and A^{*} is the only matrix with this property.

Proof. With $\langle u, v\rangle = v^* u$ from Definition 12.16,

$$\langle Ax, y\rangle = y^* Ax = (A^* y)^* x = \langle x, A^* y\rangle .$$

If B also satisfies the identity then $\langle x, (A^* - B)y\rangle = 0$ for all x and y, so $(A^* - B)y = \mathbf{0}$ for every y, whence $B = A^*$. □

Definition 14.2 (Adjoint of an operator)**.** *Let $T : V \to W$ be linear between finite-dimensional inner product spaces. The* adjoint *$T^* : W \to V$ is the unique linear map with $\langle Tv, w\rangle = \langle v, T^* w\rangle$ for all $v \in V$, $w \in W$; in orthonormal bases its matrix is the conjugate transpose of the matrix of T. We call T* self-adjoint *if $V = W$ and $T^* = T$, which for a real matrix means $A^t = A$ and for a complex one $A^* = A$.*

The rules are the ones Proposition 4.9 gave for the transpose, with one change: $(\lambda T)^* = \bar{\lambda}\, T^*$, because the inner product is conjugate-linear in its second slot. Also $(ST)^* = T^* S^*$ and $T^{**} = T$.

Example 14.3 (Self-adjoint and not)**.** *The real symmetric matrices of Chapter 4 are self-adjoint, as are the Hermitian matrices such as $\left(\begin{smallmatrix} 2 & 1-i \\ 1+i & 3 \end{smallmatrix}\right)$, whose off-diagonal entries are conjugates and whose diagonal is real. Every orthogonal projection is self-adjoint by Proposition 13.6(c). The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not, and neither is any matrix with a nonreal eigenvalue, by the next proposition.*

Proposition 14.4 (Self-adjoint operators behave well)**.** *Let T be self-adjoint on a finite-dimensional inner product space.*

(a) Every eigenvalue of T is real.

(b) Eigenvectors for distinct eigenvalues are orthogonal*, not merely independent.*

(c) $\langle Tv, v\rangle$ is real for every v.

Proof. (a) Let $Tv = \lambda v$ with $v \neq \mathbf{0}$. Then

$$\lambda \langle v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \bar{\lambda} \langle v, v\rangle ,$$

and $\langle v, v\rangle > 0$, so $\lambda = \bar{\lambda}$.

(b) Let $Tv = \lambda v$ and $Tw = \mu w$ with $\lambda \neq \mu$. Both are real by (a), so

$$\lambda \langle v, w\rangle = \langle Tv, w\rangle = \langle v, Tw\rangle = \langle v, \mu w\rangle = \mu \langle v, w\rangle ,$$

giving $(\lambda - \mu)\langle v, w\rangle = 0$ and hence $\langle v, w\rangle = 0$.

(c) $\langle Tv, v\rangle = \langle v, Tv\rangle = \overline{\langle Tv, v\rangle}$. □

Key Idea

Part (b) is a genuine strengthening of Theorem 9.13. For a general operator, eigenvectors for distinct eigenvalues are independent and can sit at any angle; for a self-adjoint one they are perpendicular. That is the whole difference between Chapter 10, where diagonalisation might fail and the basis P might be badly conditioned, and this chapter, where it never fails and the basis is orthonormal.

14.2 Schur's theorem

Theorem 14.5 (Schur). *Every $A \in M_n(\mathbb{C})$ can be written*

$$A = QTQ^*$$

with Q unitary and T upper triangular. The diagonal entries of T are the eigenvalues of A, each appearing as often as its algebraic multiplicity.

Proof. This is the induction of Lemma 11.3, run with orthonormal bases so that the change of basis comes out unitary. Induct on n; the case $n = 1$ is trivial. For $n > 1$, Theorem 9.9 gives an eigenvalue λ with an eigenvector, which we scale to a unit vector u. Extend u to an orthonormal basis of $\mathbb{C}^n$ by Theorem 12.10 and let Q_1 be the unitary matrix with those vectors as columns, u first. The first column of $Q_1^* A Q_1$ is $Q_1^* A u = \lambda\, Q_1^* u = \lambda e_1$, so

$$Q_1^* A Q_1 = \begin{pmatrix} \lambda & b^* \\ 0 & A_1 \end{pmatrix}$$

for some vector b and some A_1 of size $n-1$. By induction $A_1 = Q_2 T_1 Q_2^*$ with Q_2 unitary and T_1 upper triangular. Put $Q = Q_1 \left(\begin{smallmatrix} 1 & 0 \\ 0 & Q_2 \end{smallmatrix}\right)$, a product of unitaries and hence unitary. Then

$$Q^* A Q = \begin{pmatrix} 1 & 0 \\ 0 & Q_2^* \end{pmatrix} \begin{pmatrix} \lambda & b^* \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & Q_2 \end{pmatrix} = \begin{pmatrix} \lambda & b^* Q_2 \\ 0 & T_1 \end{pmatrix},$$

which is upper triangular. For the diagonal, T is similar to A, so they share a characteristic polynomial by Proposition 9.6, and a triangular matrix's characteristic polynomial is $\prod_i (t_{ii} - \lambda)$ by Proposition 9.18(c). □

Key Idea

Compare this with Chapter 10. Diagonalisation asks for a basis of eigenvectors and sometimes there is none. Schur asks only for an orthonormal basis in which the matrix is triangular, and there is always one, over $\mathbb{C}$. The price is the entries above the diagonal; the gain is that Q is unitary, so nothing is lost to conditioning. This is the factorisation Figure 10.3 advertised and the one the QR algorithm computes.

14.3 The spectral theorem

Theorem 14.6 (Spectral theorem, complex case)**.** *Let $A \in M_n(\mathbb{C})$ be self-adjoint. Then*

$$A = QDQ^*$$

with Q unitary and D real diagonal. Equivalently, $\mathbb{C}^n$ has an orthonormal basis of eigenvectors of A, and all the eigenvalues are real.

Proof. By Theorem 14.5, $A = QTQ^*$ with $T = Q^*AQ$ upper triangular. Then

$$T^* = (Q^*AQ)^* = Q^*A^*Q = Q^*AQ = T,$$

so T is self-adjoint as well. But a self-adjoint upper triangular matrix is diagonal: for $i < j$ we have $t_{ji} = 0$ by triangularity, and $t_{ij} = \overline{t_{ji}} = 0$ by self-adjointness. Its diagonal is real because $t_{ii} = \overline{t_{ii}}$. Taking $D = T$ finishes, and the columns of Q are then an orthonormal basis of eigenvectors, since $AQ = QD$ reads column by column as $Aq_i = d_i q_i$. □

Theorem 14.7 (Spectral theorem, real symmetric case)**.** *Let $A \in M_n(\mathbb{R})$ satisfy $A^t = A$. Then $A = QDQ^t$ with Q real orthogonal and D real diagonal. In particular $\mathbb{R}^n$ has an orthonormal basis of eigenvectors of A, so every real symmetric matrix is diagonalisable.*

Proof. The characteristic polynomial of A has a complex root by the fundamental theorem of algebra, and every root is real by Proposition 14.4(a); so A has a real eigenvalue, and $N(A - \lambda \mathbf{1})$ is a nonzero *real* subspace, which supplies a real unit eigenvector. The induction of Theorem 14.5 can therefore be run entirely over $\mathbb{R}$, with Q_1 real orthogonal, and the argument of Theorem 14.6 then forces the triangular matrix to be diagonal. □

Key Idea

That is the third promise paid. A real symmetric matrix is never defective, its eigenvalues are never complex, and its eigenvectors can always be chosen perpendicular and of unit length. Both worked examples of Chapter 10 were symmetric, which is why they came out so cleanly; it was not luck.

Two pictures make the point from different sides. Figure 14.1 sets a symmetric matrix beside one that is diagonalisable and not symmetric, where the eigenvectors are independent but not perpendicular, so the diagonalising matrix is not orthogonal. Figure 14.2 then draws the symmetric case acting on the unit circle: the two eigendirections meet at a right angle and the axes of the image ellipse lie along them.

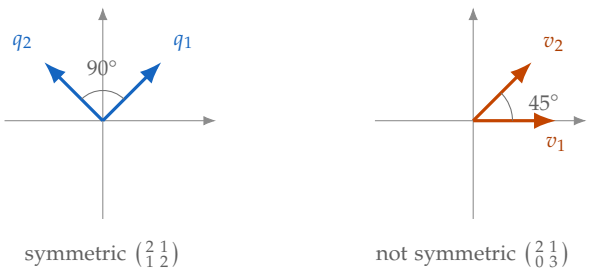


Figure 14.1. What symmetry buys. Both matrices have two real eigenvalues and two independent eigenvectors, so both are diagonalisable; the difference is the angle. The symmetric matrix has eigenvectors at 45 and 135 degrees, exactly perpendicular, so the diagonalising matrix is orthogonal and inverting it costs a transpose. The nonsymmetric one has eigenvectors only 45 degrees apart, so its diagonalising matrix is invertible but not orthogonal, and its condition number enters every perturbation bound in Chapter 31. Push the two directions closer together and that condition number diverges; in the limit they coincide, the matrix is defective, and no diagonalisation exists at all.

Example 14.8 (A symmetric diagonalisation, orthonormally)**.** *Take* $A = \left(\begin{smallmatrix}3&1\\1&3\end{smallmatrix}\right)$*, with eigenvalues* 4 *and* 2 *and eigenvectors* $(1,1)$ *and* $(1,-1)$*. Those are already orthogonal, as they must be; normalising gives*

$$Q = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad D = \begin{pmatrix}4 & 0\\ 0 & 2\end{pmatrix},$$

and Q *is orthogonal:* $Q^tQ = \frac{1}{2}\left(\begin{smallmatrix}2&0\\0&2\end{smallmatrix}\right) = \mathbf{1}$*. Note that* $Q^{-1} = Q^t$ *here, so no inverse has to be computed, which is exactly the advantage Chapter 12 advertised. Compare Example 10.4,*

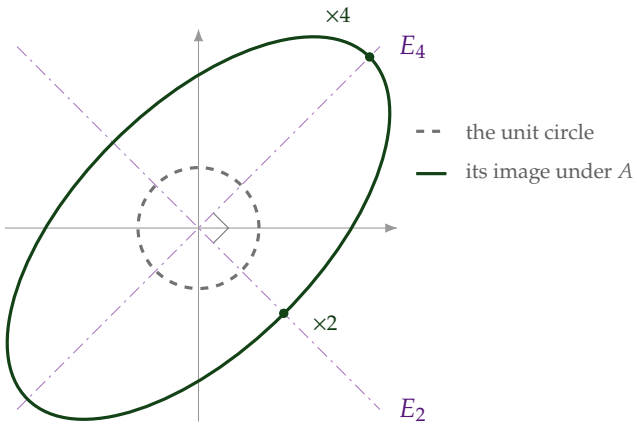


Figure 14.2. The spectral theorem, drawn for the symmetric matrix $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$. The two eigendirections meet at right angles, as Proposition 14.4(b) requires, and the image of the unit circle is an ellipse whose axes lie along them, stretched by the eigenvalues 4 and 2. For a matrix that is not symmetric this fails: the shear of Example 9.17 has only one eigendirection, and $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ has two that are not perpendicular.

where the same matrix shifted by one was diagonalised with an unnormalised P and we had to notice $P^{-1} = \frac{1}{2}P$ by hand.

14.4 Which matrices are unitarily diagonalisable?

Self-adjointness is sufficient but not necessary: a unitary matrix is unitarily diagonalisable too, and it is not self-adjoint in general. The exact condition is a small surprise.

Definition 14.9 (Normal). $A \in M_n(\mathbb{C})$ *is* normal *if* $AA^* = A^*A$.

Theorem 14.10 (Spectral theorem for normal matrices). $A \in M_n(\mathbb{C})$ *is unitarily diagonalisable if and only if it is normal.*

Proof. If $A = QDQ^*$ then $AA^* = QDD^*Q^*$ and $A^*A = QD^*DQ^*$, and diagonal matrices commute, so A is normal.

Conversely let A be normal and write $A = QTQ^*$ by Theorem 14.5. Then $T = Q^*AQ$ is normal too, since conjugating by a unitary respects both products. Now compare the $(1,1)$ entries of TT^* and T^*T: the first is $\sum_j |t_{1j}|^2$ and the second is $|t_{11}|^2$, because T is upper triangular. They are equal, so $t_{1j} = 0$ for every $j > 1$. Deleting the first row and

column leaves a smaller normal triangular matrix, and induction finishes. □

Example 14.11 (The normal family). *Self-adjoint matrices are normal, since $AA^* = A^2 = A^*A$. So are* skew-*adjoint ones, with $A^* = -A$, and unitary ones, with $AA^* = \mathbf{1} = A^*A$. The rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ is orthogonal, hence normal, hence unitarily diagonalisable over $\mathbb{C}$; its eigenvalues are $\pm i$ from Example 9.10, which are neither real nor equal, so it is certainly not self-adjoint. The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is not normal: the two products are $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$.*

Proposition 14.12 (Spectral decomposition). *Let A be normal with distinct eigenvalues $\lambda_1, \dots, \lambda_k$ and let P_i be the orthogonal projection onto E_{λ_i}. Then*

$$A = \lambda_1 P_1 + \cdots + \lambda_k P_k, \qquad P_1 + \cdots + P_k = \mathbf{1}, \qquad P_i P_j = 0 \text{ for } i \neq j.$$

Proof. Group the columns of Q from Theorem 14.10 according to which eigenvalue they belong to. The columns in group i form an orthonormal basis of E_{λ_i}, so by Proposition 13.6(a) the corresponding $Q_i Q_i^*$ is P_i. Expanding QDQ^* by these groups gives the first identity; the second is $QQ^* = \mathbf{1}$ split the same way, and the third holds because columns in different groups are orthogonal. □

14.5 Quadratic forms

Definition 14.13 (Quadratic form and definiteness). *For real symmetric A, the function $q(x) = x^t A x = \langle Ax, x \rangle$ is the associated* quadratic form. *Call A* positive definite *if $q(x) > 0$ for every $x \neq \mathbf{0}$,* positive semidefinite *if $q(x) \geq 0$ always, and* indefinite *if q takes both signs.*

Theorem 14.14 (Principal axes, and the definiteness test). *Let A be real symmetric with $A = QDQ^t$ as in Theorem 14.7. Substituting $x = Qy$,*

$$q(x) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

Hence A is positive definite exactly when every $\lambda_i > 0$, positive semidefinite exactly when every $\lambda_i \geq 0$, and indefinite exactly when the eigenvalues take both signs.

Proof. $q(Qy) = (Qy)^t A (Qy) = y^t Q^t A Q\, y = y^t D y = \sum_i \lambda_i y_i^2$. As y ranges over $\mathbb{R}^n$ so does $x = Qy$, since Q is invertible, and $y = \mathbf{0}$ only for $x = \mathbf{0}$. So the sign behaviour of q

is read off the λ_i: taking $y = e_i$ shows each λ_i is a value of q on a nonzero vector, and conversely a sum with all coefficients positive is positive. □

Figure 14.3 draws the level set $q(x) = 1$ for two forms. The semi-axis in each eigendirection is $1/\sqrt{\lambda}$, so the ellipse is shorter where the eigenvalue is larger, which is the one thing about these pictures that is easy to get backwards.

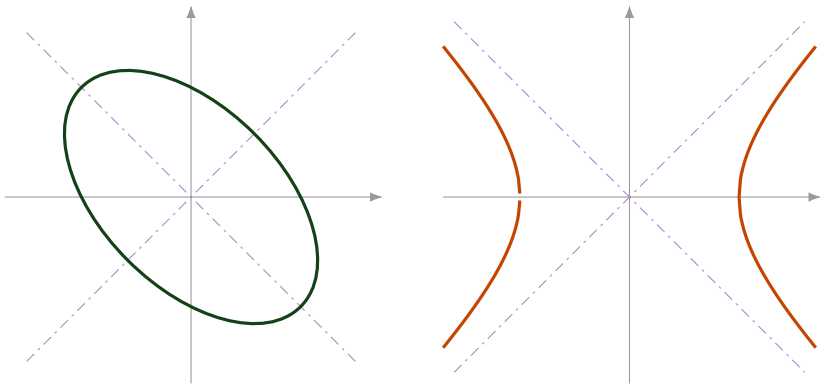


Figure 14.3. The level set $q(x) = 1$ for two quadratic forms, with the eigendirections dash-dotted. On the left $q(x) = x_1^2 + x_1x_2 + x_2^2$, whose eigenvalues are $\frac{3}{2}$ and $\frac{1}{2}$; the ellipse has semi-axis $1/\sqrt{\lambda}$ in each eigendirection, so it is *shorter* where the eigenvalue is larger. On the right $q(x) = x_1^2 - x_2^2$, where the signs disagree and the level set is unbounded. Theorem 14.14 says these two pictures, plus the degenerate cases with a zero eigenvalue, are all that can happen.

Caution

Positive definiteness is not the same as having positive entries, in either direction. $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ has all entries positive and eigenvalues 3 and -1, so it is indefinite; $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ has a negative entry and eigenvalues 3 and 1, so it is positive definite. Nor does a positive determinant suffice: $-\mathbf{1}$ has determinant 1 in even dimensions and is negative definite. The eigenvalues, or equivalently the signs of the leading principal minors, are what decide.

14.6 The Rayleigh quotient

Theorem 14.15 (Variational characterisation)**.** *Let A be real symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$. Then for $x \neq \mathbf{0}$ the* Rayleigh quotient

$$R(x) = \frac{\langle Ax, x\rangle}{\langle x, x\rangle}$$

satisfies $\lambda_n \leq R(x) \leq \lambda_1$, with $R(x) = \lambda_1$ exactly on the eigenvectors for λ_1 and $R(x) = \lambda_n$ exactly on those for λ_n.

Proof. Let $e_1, \dots, e_n$ be an orthonormal eigenbasis from Theorem 14.7 and write $x = \sum_i c_i e_i$. By Theorem 12.9,

$$\langle Ax, x\rangle = \sum_i \lambda_i c_i^2, \qquad \langle x, x\rangle = \sum_i c_i^2,$$

so $R(x)$ is a weighted average of the λ_i with weights $c_i^2 / \sum_j c_j^2$, which are nonnegative and sum to 1. A weighted average lies between the largest and smallest values, and equals the largest only when all the weight sits on indices with $\lambda_i = \lambda_1$, that is, only when $x \in E_{\lambda_1}$. □

Figure 14.4 follows the quotient once round the unit circle. It sweeps the closed interval between the two eigenvalues and never leaves it, attaining each end exactly on the corresponding eigendirection.

Insight

This is principal component analysis. Given data whose columns have been centred, the covariance matrix $C = \frac{1}{N} X X^t$ is symmetric and positive semidefinite, so Theorem 14.7 supplies an orthonormal basis of eigenvectors and Theorem 14.14 says the eigenvalues are nonnegative. The variance of the data projected onto a unit direction u is exactly $\langle Cu, u\rangle$, so Theorem 14.15 says the direction of greatest variance is the top eigenvector and the variance along it is the top eigenvalue. The usual definition of PCA, find the direction of maximum variance, then the best direction perpendicular to it, and so on, is therefore not a recipe that happens to work: it is the spectral theorem, and the perpendicularity that PCA assumes is Proposition 14.4(b).

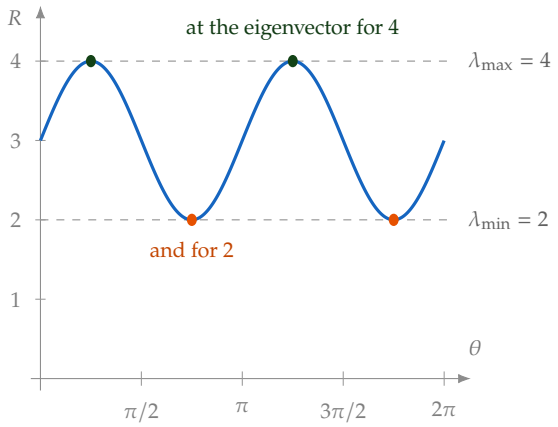


Figure 14.4. Theorem 14.15 for $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, with $x = (\cos\theta, \sin\theta)$ running once round the unit circle. Here $R(\theta) = 3 + \sin 2\theta$, which sweeps the interval between the two eigenvalues and no more. The maxima occur at $\theta = \pi/4$ and $\theta = 5\pi/4$, the two unit vectors along E_4; the minima at $3\pi/4$ and $7\pi/4$, along E_2.

Open Problem 14.16 (How close are almost commuting Hermitian matrices to commuting ones?)**.** *The spectral theorem diagonalises one Hermitian matrix, and two Hermitian matrices are simultaneously diagonalisable exactly when they commute. Suppose they nearly commute:* $\|AB - BA\| \le \varepsilon$ *with* $\|A\|, \|B\| \le 1$. Must there be commuting Hermitian A', B' with $\|A - A'\|$ and $\|B - B'\|$ small, by an amount depending on ε alone and not on the size of the matrices?

What is known. *Yes, and that is Lin's theorem, proved in 1995 and given a short proof by Friis and Rørdam [236]. But the proof is by a compactness and C*-algebra argument that produces no rate at all: it shows a function* $\delta(\varepsilon)$ *tending to zero exists, without saying whether it is* $\sqrt{\varepsilon}$ *or something far worse.* No explicit bound is known. *The best available constructions show* $\delta(\varepsilon)$ *cannot be better than about* $\varepsilon^{1/2}$ *up to constants, and the true rate is open. For three Hermitian matrices the statement is simply false, by an obstruction of Voiculescu that is topological in nature and can be exhibited by an index computation [305].*

Why two and not three. *The three-matrix counterexample is built from matrices that behave like coordinates on a sphere, and the obstruction is that a certain vector bundle over the sphere is nontrivial. There is no such obstruction over a disc, which is why two matrices are safe. This is one of the few places in matrix analysis where topology decides the answer, and it means any quantitative proof of Lin's theorem must be sensitive to why the two-dimensional case is topologically trivial.*

Where a reader might start. *Construct almost commuting Hermitian matri-*

ces that are not obviously *near commuting ones: the clock and shift matrices* $C = \operatorname{diag}(1, \omega, \dots, \omega^{n-1})$ *and the cyclic shift* S*, with* $\omega = e^{2\pi i/n}$*, satisfy* $CS = \omega SC$*, so their commutator is* $O(1/n)$ *after taking Hermitian and anti-Hermitian parts. Then find the nearby commuting pair explicitly and measure the distance as a function of* n*.*

Exercises

Exercise 14.17. *Diagonalise each symmetric matrix orthogonally, that is, find a real orthogonal* Q *and diagonal* D *with* $A = QDQ^{t}$*.*

(a) $\left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right)$

(b) $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$

(c) $\left(\begin{smallmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right)$

Exercise 14.18. *Verify* $\langle Ax, y \rangle = \langle x, A^{*}y \rangle$ *directly for* $A = \left(\begin{smallmatrix} 1 & i \\ 0 & 2 \end{smallmatrix}\right)$*,* $x = (1, i)$ *and* $y = (i, 1)$ *in* $\mathbb{C}^2$*. Is* A *normal?*

Exercise 14.19. *Classify each form as positive definite, positive semidefinite, negative definite or indefinite, by computing eigenvalues.*

(a) $q(x) = 2x_1^2 + 2x_1x_2 + 2x_2^2$

(b) $q(x) = x_1^2 + 4x_1x_2 + x_2^2$

(c) $q(x) = x_1^2 + 2x_1x_2 + x_2^2$

Exercise 14.20. *Show that* $A^{t}A$ *is symmetric and positive semidefinite for every real* A*, and that it is positive definite exactly when the columns of* A *are independent. Compare Lemma 13.5.*

Exercise 14.21. *Let* A *be symmetric positive semidefinite. Show that there is a symmetric positive semidefinite* S *with* $S^2 = A$*, and that it is unique.* Hint: use Theorem 14.7 and take square roots on the diagonal; for uniqueness, note that S must act on each eigenspace of A by a scalar.

Exercise 14.22. *Compute a Schur factorisation of* $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$*, and explain why no unitary* Q *can make* $Q^{*}AQ$ *diagonal for this* A*. Which hypothesis of Theorem 14.10 fails?*

Exercise 14.23. *For* $A = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right)$ *compute* $R(x)$ *at* $x = (1,0)$*,* $(0,1)$ *and* $(1,1)$*, and check that all three values lie between the eigenvalues you found in Exercise 14.17(a). Which unit vectors attain the extremes?*

Exercise 14.24. *Let* A *and* B *be real symmetric with* $AB = BA$*. Show that they have a common orthonormal eigenbasis.* Hint: combine Exercise 10.18 with Proposition 14.4(b), applying the spectral theorem to B restricted to each eigenspace of A.

Exercise 14.25. *Let* A *be real with* $A^{t} = -A$*. Show that* $\langle Ax, x\rangle = 0$ *for every* x*, that every eigenvalue of* A *is purely imaginary, and that* A *is normal. What does Theorem 14.10 then say, and why does it not contradict the fact that a real skew matrix of odd size is singular?*

References and further reading

The spectral theorem has more good expositions than any other result in the subject, and it is worth reading several. Axler [37] proves it without determinants and without Schur, by an induction on invariant subspaces, and his account makes clearest why self-adjointness is the right hypothesis. Our route through Schur's triangularisation is the one that generalises furthest: it costs nothing extra to reach normal matrices, and it is the finite-dimensional shadow of how the theorem is proved for operators on a Hilbert space, where triangular forms survive and eigenvectors may not. Banica [51], our baseline, organises this material around exactly that observation and is the place to go next; the later parts of that book are essentially a study of what the spectral theorem becomes when the matrices are replaced by operators and the sums by integrals. Treil [610] is good on quadratic forms and on Sylvester's law of inertia, which we have only mentioned. For the numerical side, Trefethen and Bau [607] explain why the symmetric eigenvalue problem is so much better behaved than the general one, which is the computational counterpart of everything in this chapter.